

Joseph Flammia

Lab 22

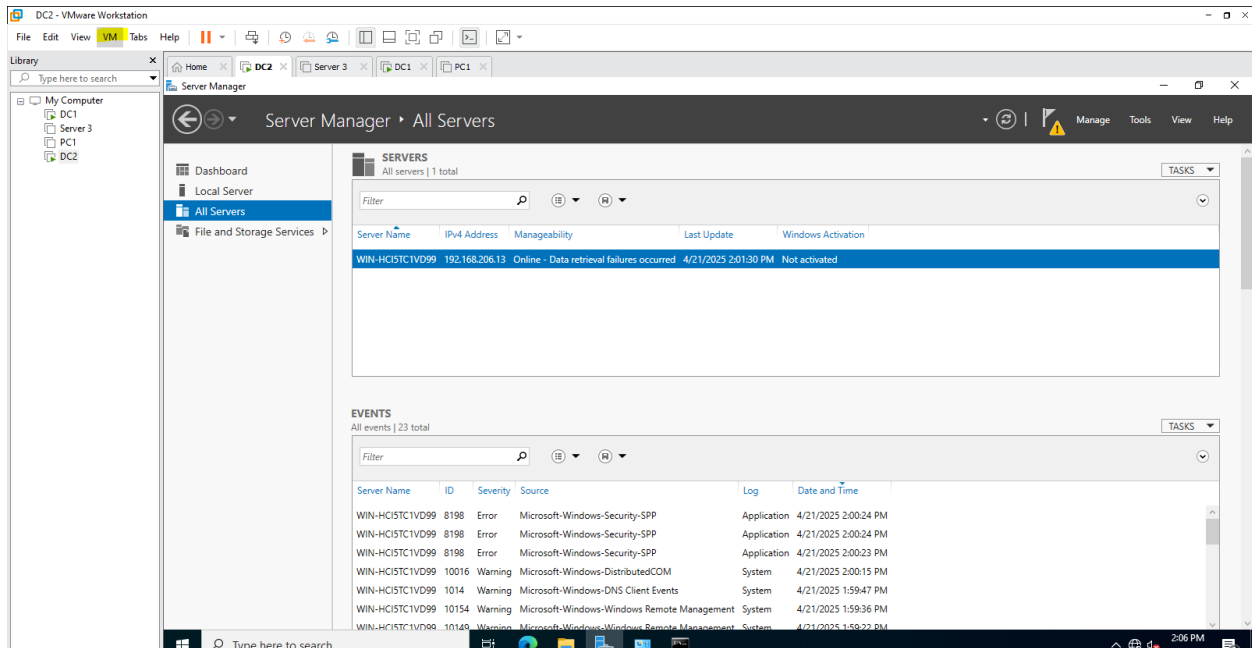
4/21/2025

Lab 22A:

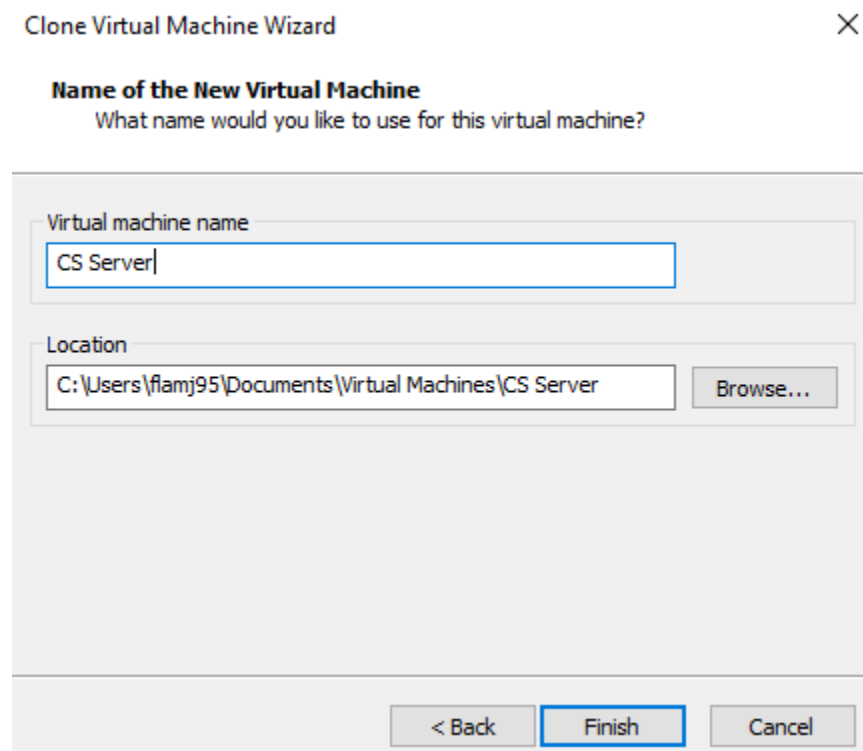
To start off, we will create a 4th server in our domain that will become a certificate server.

To do this, I created a clone of “SERVER2” by clicking on “VM”, “Manage”, and then “Clone” in VMWare.

NOTE: I cloned SERVER2 because it was an empty server with nothing installed on it (no AD, no DNS, no IIS, etc). When cloning a VM, make sure to clone one that does not have any roles installed on it, otherwise errors may occur.



For clarification purposes, I will name this server “CS Server” .



Clone Virtual Machine Wizard

Name of the New Virtual Machine
What name would you like to use for this virtual machine?

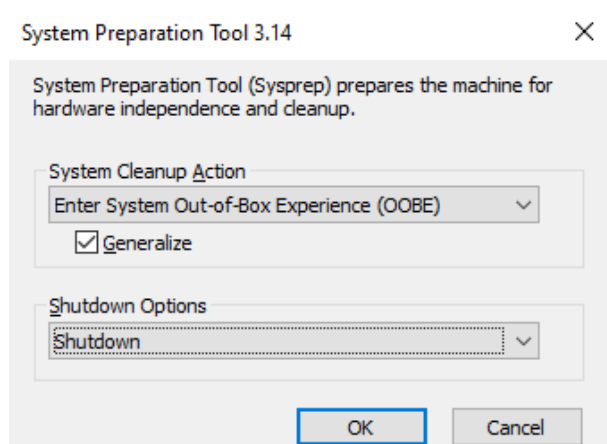
Virtual machine name
CS Server

Location
C:\Users\flamj95\Documents\Virtual Machines\CS Server Browse...

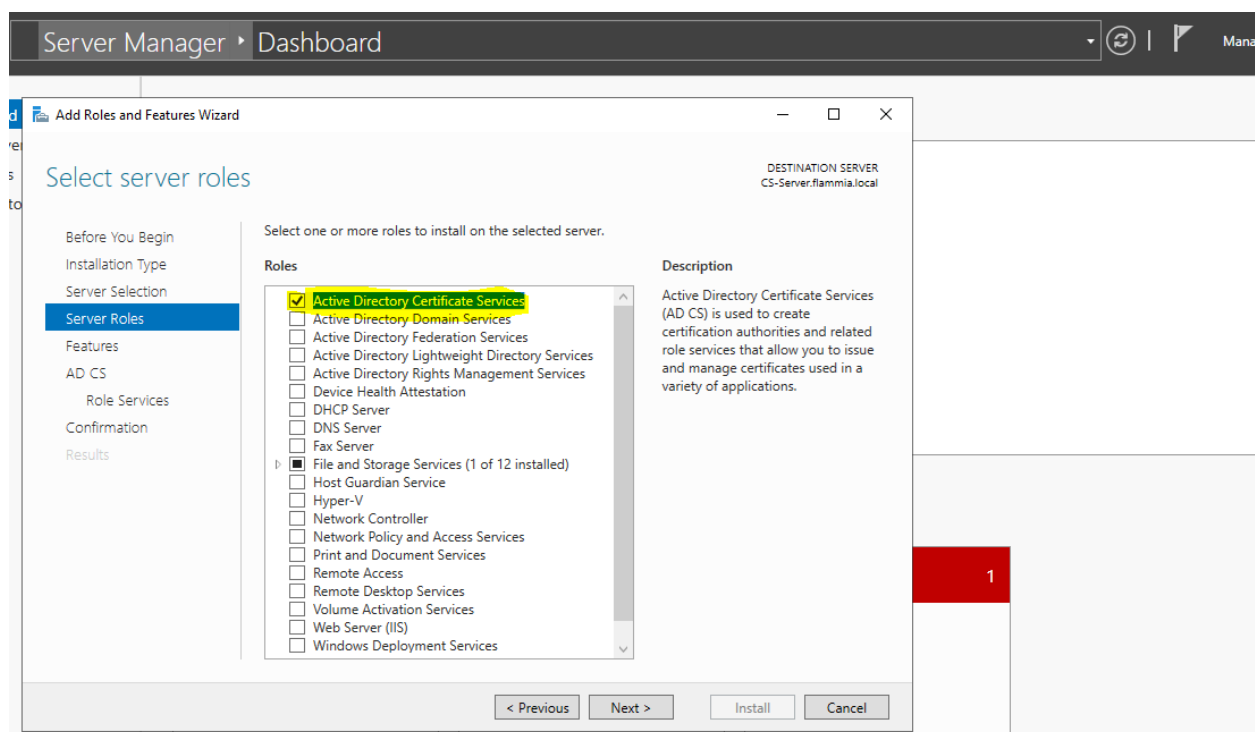
< Back Finish Cancel

Once our VM has been cloned, we will need to run a sysprep in order to get a new SID. To do this, open file explorer, and navigate to the “sysprep” application located in the sysprep file.

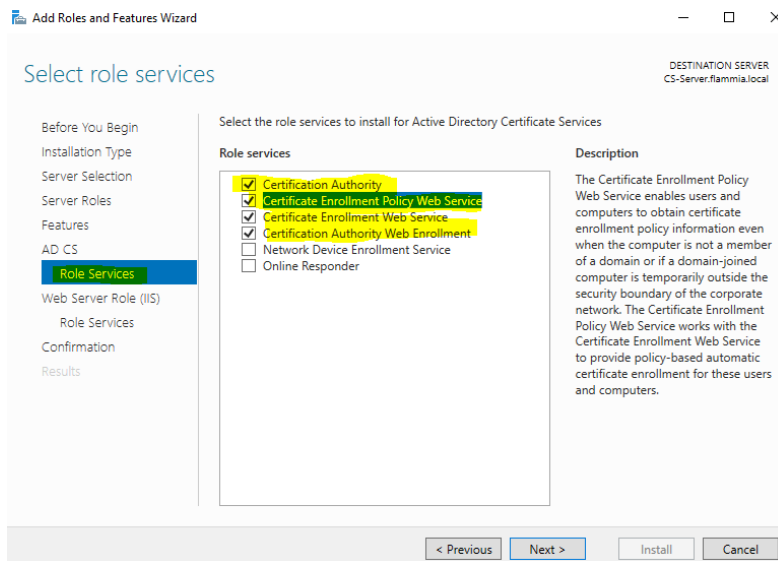
Once opened, we will need to make sure the system cleanup action is set to “OOBE”, check generalize and then set the shutdown options to “shutdown”. Once all of that is done, I clicked “OK”.



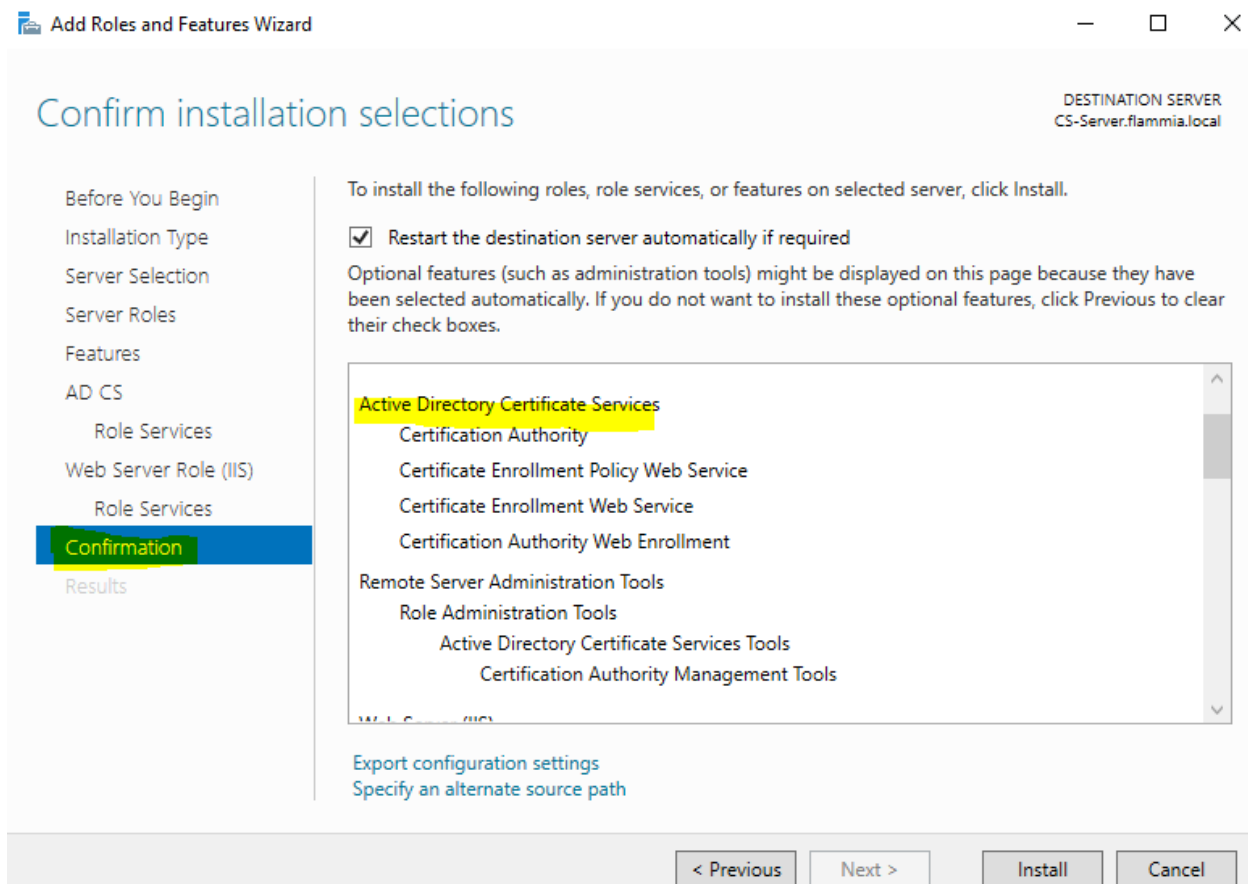
After running the sysprep, our machine is ready for use. Next up, we will install AD CS on our newly installed VM. To do this, I clicked on the top right button that says “manage” and then clicked “Add roles and features”. From the installation wizard, I checked off the “Active Directory Certificate Services” box.



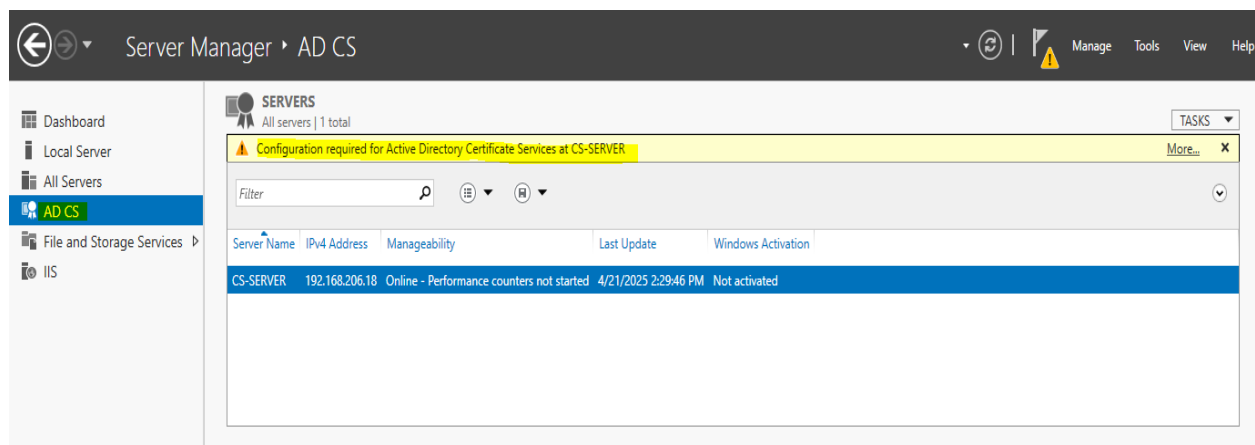
The features tab can be left alone for now, in the “Role Services” tab, check off the first four boxes to install. Depending on the system, you may need to install other services in order to utilize some of the services, such as IIS.



If everything looks good in the “Confirmation” tab, we can start the installation.



Once AD CS is installed, we can exit the wizard and head back to the server manager dashboard. On the left-hand side of the dashboard, click on” AD CS”. As you can see, some additional configurations may be required before we can continue. To do this, simply click on the “more” text on the right side.



From here, all we need to do is click on “Configure Active Directory Certificates” as highlighted below.

All Servers Task Details

All Servers Task Details and Notifications

All Tasks | 1 total

Filter

Status	Task Name	Stage	Message	Action	Notifications
	Post-deployment Configuration	Not Sta...	Configuration required for Active Directory Cer...	Configure Active Directory Certi...	1

Status	Notification	Time Stamp
	Additional steps are required to configure Active Directory Certificate Services on the destination server	4/21/2025 2:29:42 PM

From this window, we can skip the credentials screen and head over to the “role services” tab. For now, all we will have to do is install the “Certification Authority” role. We will come back and install the other roles later. After checking of CA, click next.

Role Services

DESTINATION SERVER
CS-Server.flammia.local

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

Select Role Services to configure

- ☒ Certification Authority
- ☐ Certification Authority Web Enrollment
- ☐ Online Responder
- ☐ Network Device Enrollment Service
- ☐ Certificate Enrollment Web Service
- ☐ Certificate Enrollment Policy Web Service

[More about AD CS Server Roles](#)

< Previous

Next >

Configure

Cancel

In the “setup type” tab, since we are a member of our domain, ensure that “Enterprise CA” is checked off. Once checked off, click next.

Setup Type

DESTINATION SERVER
CS-Server.flammia.local

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

Specify the setup type of the CA

Enterprise certification authorities (CAs) can use Active Directory Domain Services (AD DS) to simplify the management of certificates. Standalone CAs do not use AD DS to issue or manage certificates.

☒ Enterprise CA

Enterprise CAs must be domain members and are typically online to issue certificates or certificate policies.

☐ Standalone CA

Standalone CAs can be members or a workgroup or domain. Standalone CAs do not require AD DS and can be used without a network connection (offline).

[More about Setup Type](#)

< Previous

Next >

Configure

Cancel

Next up is the CA type. Since this is the first CA in our domain we will choose “Root CA”. Once done, hit next.

CA Type

DESTINATION SERVER
CS-Server.flammia.local

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

Specify the type of the CA

When you install Active Directory Certificate Services (AD CS), you are creating or extending a public key infrastructure (PKI) hierarchy. A root CA is at the top of the PKI hierarchy and issues its own self-signed certificate. A subordinate CA receives a certificate from the CA above it in the PKI hierarchy.

☒ Root CA

Root CAs are the first and may be the only CAs configured in a PKI hierarchy.

☐ Subordinate CA

Subordinate CAs require an established PKI hierarchy and are authorized to issue certificates by the CA above them in the hierarchy.

[More about CA Type](#)

< Previous

Next >

Configure

Cancel

On the private key tab, we will want to create a new private key since this is our first CA. Once checked off, click next.

AD CS Configuration

DESTINATION SERVER
CS-Server.flammia.local

Private Key

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key**
- Cryptography
- CA Name
- Validity Period
- Certificate Database
- Confirmation
- Progress
- Results

Specify the type of the private key

To generate and issue certificates to clients, a certification authority (CA) must have a private key.

- ☒ **Create a new private key**
Use this option if you do not have a private key or want to create a new private key.
- ☐ Use existing private key
Use this option to ensure continuity with previously issued certificates when reinstalling a CA.
 - ☐ Select a certificate and use its associated private key
Select this option if you have an existing certificate on this computer or if you want to import a certificate and use its associated private key.
 - ☐ Select an existing private key on this computer
Select this option if you have retained private keys from a previous installation or want to use a private key from an alternate source.

[More about Private Key](#)

< Previous Next > Configure Cancel

For the “Cryptography” section, we can leave the defaults as is. Now for the “CA Name” tab, this is what our certificate authority will be identified as. In my case, I left the default name, but you can make this whatever you would like.

AD CS Configuration

DESTINATION SERVER
CS-Server.flammia.local

CA Name

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key
- Cryptography
- CA Name**
- Validity Period
- Certificate Database
- Confirmation
- Progress
- Results

Specify the name of the CA

Type a common name to identify this certification authority (CA). This name is added to all certificates issued by the CA. Distinguished name suffix values are automatically generated but can be modified.

Common name for this CA:

Distinguished name suffix:

Preview of distinguished name:

[More about CA Name](#)

< Previous Next > Configure Cancel

For the validity period, I set the validity of issued certificates to 5 years, this can be changed to whatever suits your preference, however.

AD CS Configuration

DESTINATION SERVER
CS-Server.flammia.local

Validity Period

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key
- Cryptography
- CA Name
- Validity Period**
- Certificate Database
- Confirmation
- Progress
- Results

Specify the validity period

Select the validity period for the certificate generated for this certification authority (CA):

CA expiration Date: 4/21/2030 2:34:00 PM

The validity period configured for this CA certificate should exceed the validity period for the certificates it will issue.

[More about Validity Period](#)

< Previous Next > Configure Cancel

Finally, for the “Certificate Database” tab, we can leave this to the default file locations. After that, we can check the confirmation tab to ensure all of our configurations are correct. If everything looks good, click on configure to continue.

The screenshot shows the 'AD CS Configuration' window with the 'Confirmation' tab selected. The window title is 'AD CS Configuration'. In the top right corner, it says 'DESTINATION SERVER CS-Server.flammia.local'. On the left, a navigation pane lists: Credentials, Role Services, Setup Type, CA Type, Private Key, Cryptography, CA Name, Validity Period, Certificate Database, Confirmation (highlighted), Progress, and Results. The main area has a heading 'Active Directory Certificate Services' with a sub-heading 'Certification Authority'. Below this, a list of configuration details is shown: CA Type: Enterprise Root, Cryptographic provider: RSA#Microsoft Software Key Storage Provider, Hash Algorithm: SHA256, Key Length: 2048, Allow Administrator Interaction: Disabled, Certificate Validity Period: 4/21/2030 2:34:00 PM, Distinguished Name: CN=flammia-CS-SERVER-CA,DC=flammia,DC=local, Certificate Database Location: C:\Windows\system32\CertLog, Certificate Database Log Location: C:\Windows\system32\CertLog. At the bottom, there are four buttons: '< Previous', 'Next >', 'Configure', and 'Cancel'.

Certification Authority	
CA Type:	Enterprise Root
Cryptographic provider:	RSA#Microsoft Software Key Storage Provider
Hash Algorithm:	SHA256
Key Length:	2048
Allow Administrator Interaction:	Disabled
Certificate Validity Period:	4/21/2030 2:34:00 PM
Distinguished Name:	CN=flammia-CS-SERVER-CA,DC=flammia,DC=local
Certificate Database Location:	C:\Windows\system32\CertLog
Certificate Database Log Location:	C:\Windows\system32\CertLog

If our configurations were successful, we should see a “Configuration Succeeded” message. From here, we can close the configuration window and now proceed to configure the remaining roles as mentioned before.

Results

DESTINATION SERVER
CS-Server.flammia.local

Credentials
Role Services
Setup Type
CA Type
Private Key
 Cryptography
 CA Name
 Validity Period
Certificate Database
Confirmation
Progress
Results

The following roles, role services, or features were configured:

Active Directory Certificate Services

Certification Authority

[More about CA Configuration](#)

✓ Configuration succeeded

< Previous

Next >

Close

Cancel

Once back in the configuration window, we will install the following role services: “CA Web Enrollment”, “CE Web Service”, and “CE Policy Web Service”.

Role Services

Credentials
Role Services
CA for CES
Authentication Type for C...
Service Account for CES
Authentication Type for C...
Server Certificate
Confirmation
Progress
Results

Select Role Services to configure

- ☒ Certification Authority
- ☒ Certification Authority Web Enrollment
- ☐ Online Responder
- ☐ Network Device Enrollment Service
- ☒ Certificate Enrollment Web Service
- ☒ Certificate Enrollment Policy Web Service

For most of the configurations, the defaults should work just fine. However, for “Service Account for CES”, checking the “built-in application pool identity” will allow for an easier time later on.

 AD CS Configuration— □ ×

Service Account for CES

Credentials

Role Services

CA for CES

Authentication Type for C...

Service Account for CES

Authentication Type for C...

Server Certificate

Confirmation

Progress

Results

Specify the service account

Select the identity that the Certificate Enrollment Web Service (CES) uses when communicating with the certification authority (CA) and other services on the network.

☐ Specify service account (recommended)

The account selected must be a member of the IIS_IUSRS group. If Kerberos is selected as the authentication type, a service principal name is required for the service account.

☒ Use the built-in application pool identity

[More about Service Account for CES](#)

DESTINATION SERVER

CS-Server.flammia.local

< Previous

Next >

Configure

Cancel

Now in the “server certificate” tab, we will need to choose the certificate that was issued to the server when configuring the CA before.

The screenshot shows the 'AD CS Configuration' console window. The 'Server Certificate' tab is selected in the left-hand navigation pane. The main area is titled 'Specify a Server Authentication Certificate'. It contains a description of SSL encryption and two radio button options. The first option, 'Choose an existing certificate for SSL encryption (recommended)', is selected. Below it is a table with three columns: 'Issued To', 'Issued By', and 'Expiration Date'. A single row is visible, showing 'flammia-CS-SERVER-CA' for both 'Issued To' and 'Issued By', and '4/21/2030' for 'Expiration Date'. Below the table are 'Properties' and 'Refresh' buttons. The second radio button option is 'Choose and assign a certificate for SSL later', followed by a warning icon and a message: 'For this role service to function, you must configure this server with a valid certificate.' At the bottom of the console, there are four buttons: '< Previous', 'Next >', 'Configure', and 'Cancel'.

AD CS Configuration

DESTINATION SERVER
CS-Server.flammia.local

Server Certificate

- Credentials
- Role Services
- CA for CES
- Authentication Type for C...
- Service Account for CES
- Authentication Type for C...
- Server Certificate**
- Confirmation
- Progress
- Results

Specify a Server Authentication Certificate

When communicating with clients, the web service(s) uses Secure Sockets Layer (SSL) protocol to encrypt network traffic.

☒ Choose an existing certificate for SSL encryption (recommended)

Issued To	Issued By	Expiration Date
flammia-CS-SERVER-CA	flammia-CS-SERVER-CA	4/21/2030

☐ Choose and assign a certificate for SSL later

⚠ For this role service to function, you must configure this server with a valid certificate.

[More about Server Certificate](#)

< Previous Next > Configure Cancel

After selecting the right certificate, we can confirm all our changes and configurations are correct in the confirmation tab. If everything checks out, we can click “Configure”.

AD CS Configuration

Confirmation

Credentials

Role Services

CA for CES

Authentication Type for C...

Service Account for CES

Authentication Type for C...

Server Certificate

Confirmation

Progress

Results

DESTINATION SERVER

CS-Server.flammia.local

To configure the following roles, role services, or features, click Configure.

Active Directory Certificate Services

Certification Authority Web Enrollment

Certificate Enrollment Web Service

CA Name: CS-Server.flammia.local\flammia-CS-SERVER-CA

Renewal Only Mode: False

Authentication Type: Windows Integrated Authentication

Allow Key-based Renewal: False

Account: Application Pool Identity

Server Authentication Certificate: 943CD84759C9BD1C556F0435DB3AF2C94B2EFA68

Certificate Enrollment Policy Web Service

Authentication Type: Windows Integrated Authentication

Enable Key-based Renewal: False

Server Authentication Certificate: 943CD84759C9BD1C556F0435DB3AF2C94B2EFA68

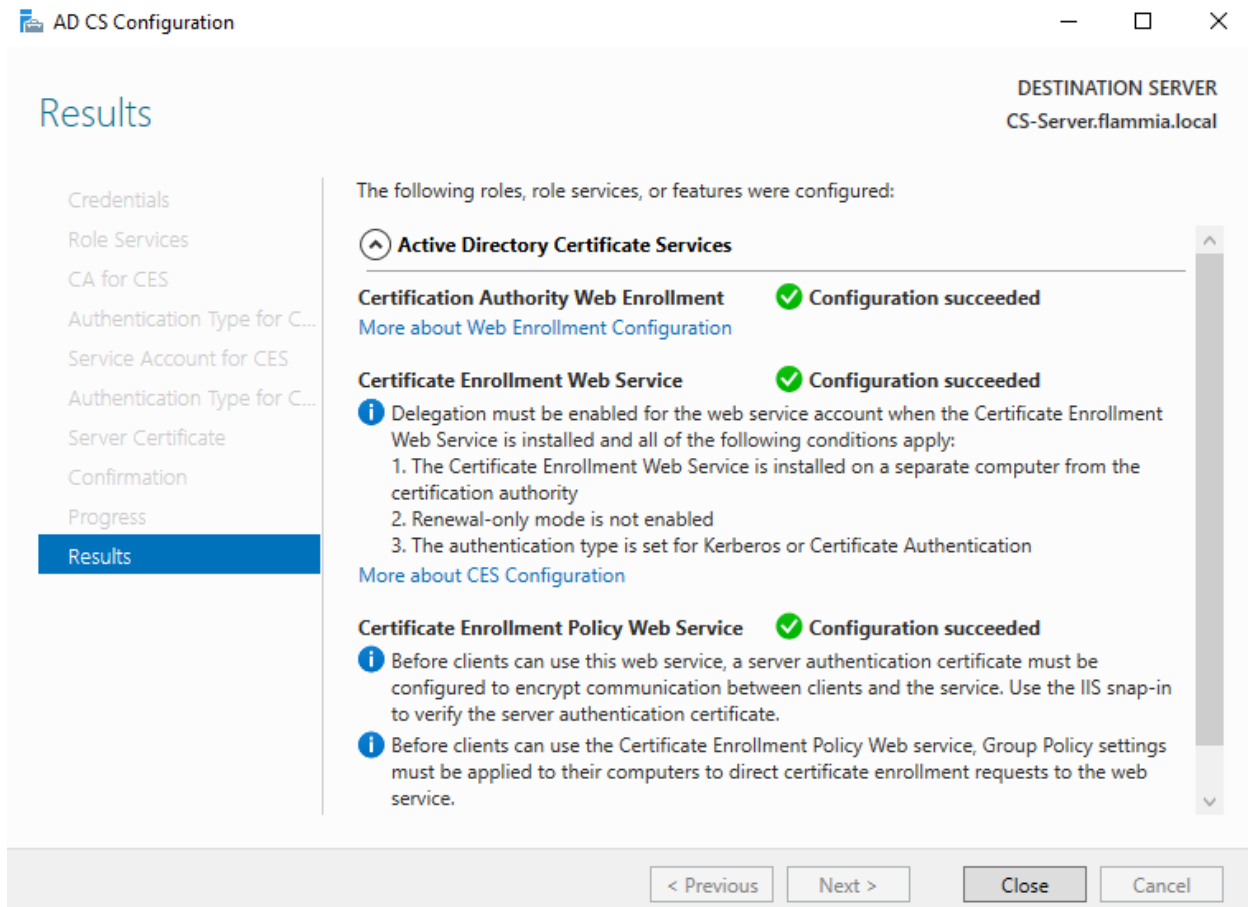
< Previous

Next >

Configure

Cancel

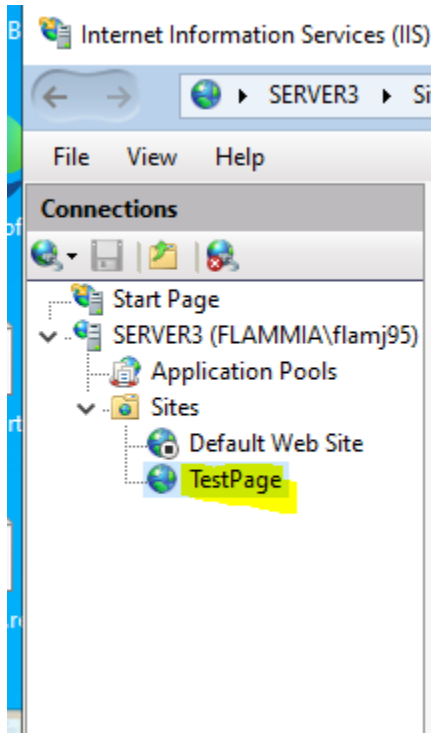
If everything was configured correctly, we should see a green checkmark for all 3 role services as pictured below:



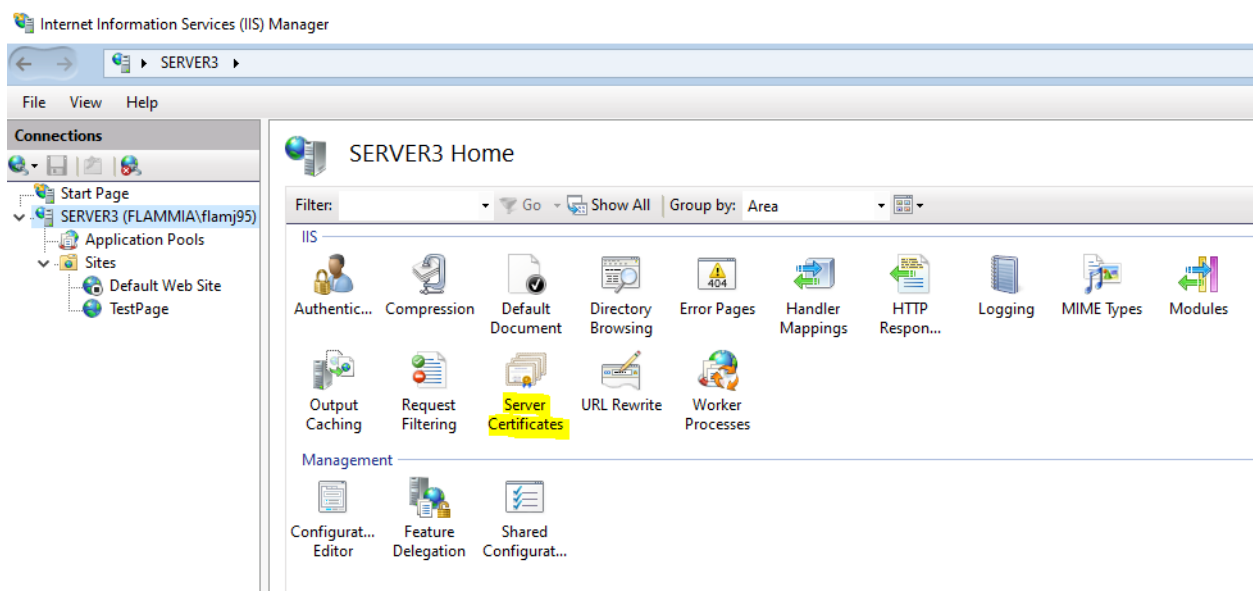
At this point, CA has now been successfully installed, and our CA Server is ready for use.

Now, we are going to switch over to our IIS server (server 3) to use a certificate issued from our CS server. For this part, I will create a new web page to start off.

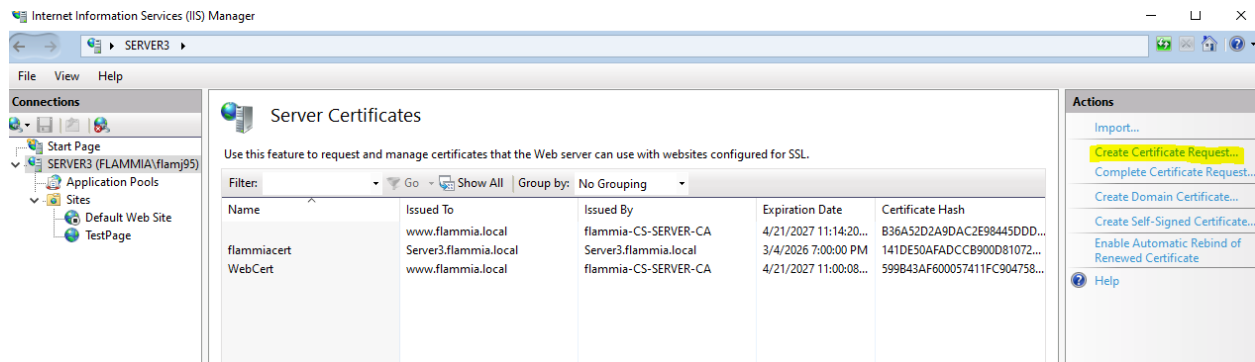
As pictured below, I created a new website on IIS called “TestPage”.



Now in the IIS manager, click on the home page of the application and then select “Server Certificates”.



From here, we can see all of the certificates that our server has issued. Next, we will navigate to the right-hand side of the page and click “Create Certificate Request”.



From here, the only critical piece of information we need to enter is the “common name”. This is the DNS record that we will use our certificate on from the CS server.

Request Certificate

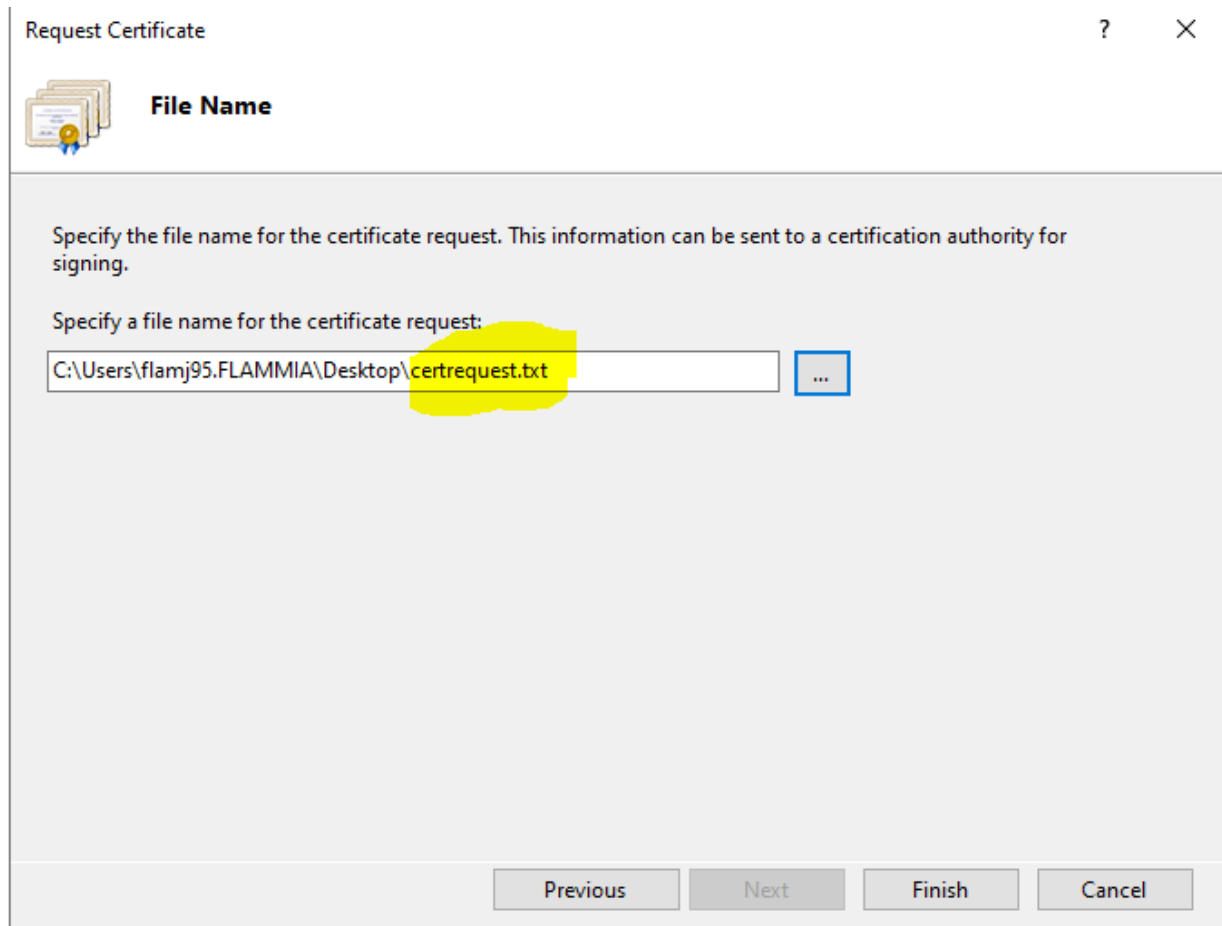
Distinguished Name Properties

Specify the required information for the certificate. State/province and City/locality must be specified as official names and they cannot contain abbreviations.

Common name:	www.flammia.local
Organization:	1
Organizational unit:	1
City/locality	1
State/province:	1
Country/region:	US

Previous Next Finish Cancel

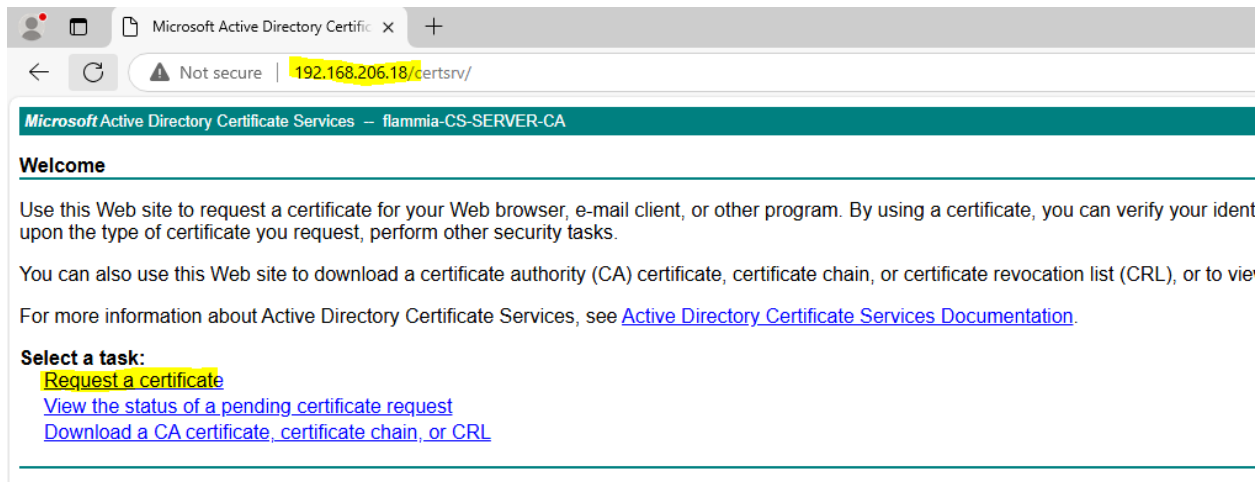
The rest of the certificate wizard's options can be just the default options. However, we will need to specify a file name for the certificate request. As pictured below, I named the file "certrequest" and designated the location on my desktop.



The screenshot shows the 'Request Certificate' wizard window. The title bar says 'Request Certificate' with a help icon and a close button. Below the title bar is a section with a certificate icon and the heading 'File Name'. The main area contains the text: 'Specify the file name for the certificate request. This information can be sent to a certification authority for signing.' followed by 'Specify a file name for the certificate request:'. Below this is a text input field containing 'C:\Users\flamj95.FLAMMIA\Desktop\certrequest.txt' and a blue button with three dots. At the bottom are four buttons: 'Previous', 'Next', 'Finish', and 'Cancel'.

Once the request has been created, we will now open a browser and enter "**192.168.206.18**/certsrv". The IP address will of course differ system to system, but it should match the IP address of the server that has AD CS installed.

If done correctly, you will be greeted with the AD CS certificate website. From here, click on "Request a certificate".



Once on the next page, we will need to copy and paste everything from the “certrequest” file into the “Saved Request” box. Furthermore, the certificate template will need to be listed as “Web Server”. As pictured below, I entered my certificate request, changed the template, and then hit submit.

Submit a Certificate Request or Renewal Request

To submit a saved request to the CA, paste a base-64-encoded CMC or PKCS #10 certificate request or PKCS #7 renewal request

Saved Request:

Base-64-encoded certificate request (CMC or PKCS #10 or PKCS #7):

```
BBYEFNgNUvILVdtbe5vnAexqCVraXXaMfA8GCS-  
6u+2sQhiiuQoI2T3cjvnt/TXis7bQaNVqQe6P3-  
2LPA/ofvpZt21rZaRP8yRMwalG+nwzK1HmB9Iy  
/gYx1KEN+9VG+kd+Ec4wThreW/v15fKFVEJ  
-----END NEW CERTIFICATE REQUEST-----
```

Certificate Template:

Web Server

Additional Attributes:

Attributes:

Submit >

After clicking submit, the certificate has now been successfully generated. From the next page, we can download the certificate as highlighted below.

Certificate Issued

The certificate you requested was issued to you.

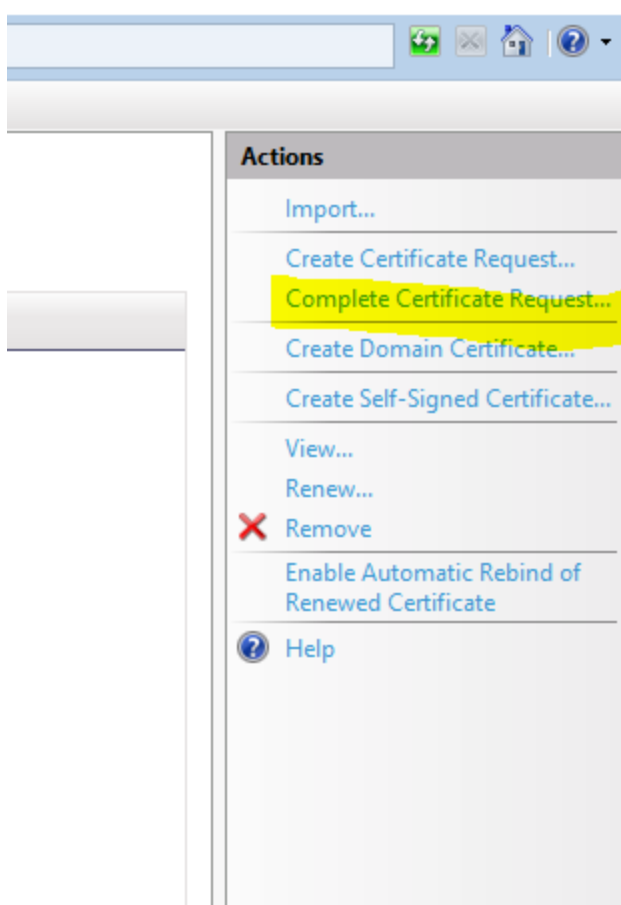
☒ DER encoded or ☐ Base 64 encoded



[Download certificate](#)

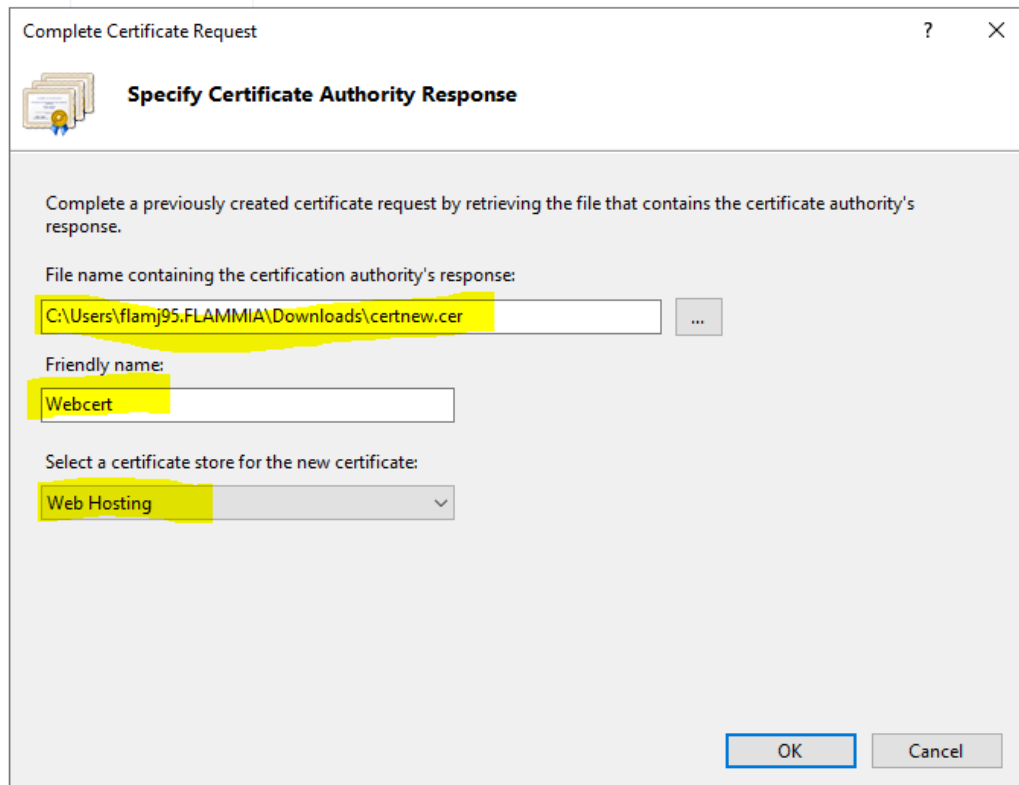
[Download certificate chain](#)

After the certificate was downloaded, head back to the IIS manager. From the home page, click on the certificates option, and then click on “Complete Certificate Request” located on the right-hand side of the application.



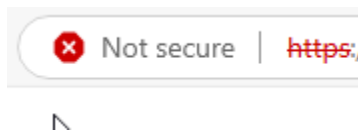
From here, a window will open up prompting us to complete the certificate request. All we need to do is specify the downloaded certificate's location, a name for the certificate, and

then set the certificate store as “Web Hosting”.



If everything was done correctly, we should now be able to access the website without any warnings.

However, you might notice that despite our configurations, the site still ends up being labeled as “insecure” and still has the same warnings as before.



Additionally, we end up getting the “ERR_CERT_COMMON_NAME_INVALID” error, despite our common name being set up correctly www.flammia.local”.

This is since most modern browsers such as Edge, Firefox, and Chrome not only require the CN, but also require a SAN (Subject Alternative Name). To proceed, we will need to configure this as well.

To start, on my IIS server I created a new document and entered the following block of text in:

```
[Version] Signature="$Windows NT$"
```

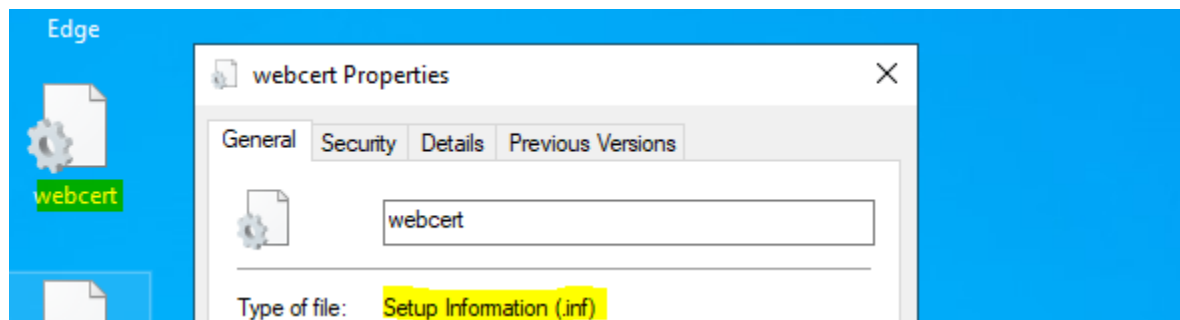
```
[NewRequest] Subject = "CN=www.flammia.local" KeySpec = 1 KeyLength = 2048  
Exportable = TRUE MachineKeySet = TRUE SMIME = FALSE PrivateKeyArchive = FALSE  
UserProtected = FALSE UseExistingKeySet = FALSE ProviderName = "Microsoft RSA  
SChannel Cryptographic Provider" ProviderType = 12 RequestType = PKCS10  
KeyUsage = 0xa0
```

```
[Extensions] 2.5.29.17 = "{text}" continue = "dns=www.flammia.local&" continue =  
"dns=flammia.local"
```

```
[RequestAttributes] CertificateTemplate = WebServer
```

The DNS record will vary from system to system, but in my case, it is "www.flammia.local".

Once entered, close the notepad file and save it as an ".inf" file.

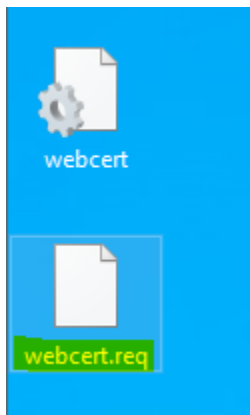


Next up, we will create the certificate request via the command “certreq -new webcert.inf webcert.req”.

NOTE: Make sure you are in the directory where the .inf file was saved. In my scenario, I saved it to the desktop.

```
C:\>cd C:\Users\flamj95.FLAMMIA\Desktop  
  
C:\Users\flamj95.FLAMMIA\Desktop>certreq -new webcert.inf webcert.req  
Active Directory Enrollment Policy  
{8B5D65D8-E3E2-44E1-89FD-1484BAE264C7}  
ldap:  
  
CertReq: Request Created
```

If done correctly, this command should create a certificate request file as pictured below:



From here, we will need to navigate back to the AD CS website. The process here will be nearly identical to before.

As pictured below, I copy and pasted the contents of the certificate request file into the Saved Request portion of the site, and then hit submit.

Microsoft Active Directory Certificate Services — flammia-CS-SERVER-CA

Submit a Certificate Request or Renewal Request

To submit a saved request to the CA, paste a base-64-encoded certificate request (CMC or PKCS #10 or PKCS #7):

Saved Request:

Base-64-encoded certificate request (CMC or PKCS #10 or PKCS #7):

-----BEGIN NEW CERTIFICATE REQUEST-----
MIIEMDCCAxgCAQAwHDEaMBGGA1UEAwRd3d3LmZsYW1tawEubG9jYmVwggEiMA0G
CSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQCWurwmiMeFODbIbkuMDZD31UjmvQ41
G8wc+amSx4+7TI870ns3y6kZ091vdRs//2sNPFP8ePKmNcJmkpftZbF5jPPP8ss
EK5r9/NcN040FF19oamgdR2hI5b2JFaUqVAmNmeXakB5mIIWd5qa++QYXn6VTXs
hg0re1LsflPS90EwEe2dFVV+WX3knTrgqG1suwGBk8/hcSMQf8cz01Pkr71te6jd
BY1AXSSp2M1pIHvBAI3T4vKx9ar0a5jzp3VUfy6EorU3uMgJ1CVPyLPoQrFY7LX7
A4DVj6G8/rIY7hiPbexDMZCeuAhArPAEnXHjyAVUhIKjy1SeJkWN8/G1AgMBAAGg
gghNMBwGC1sGAQQBgjcNAgMxMhYMTAuMC4yMDM0OC4yMEMGCSsGAQQBgjcVFDE2
MDQCAQKMFVN1cnZ1c1jMuZmxhbW1pYS5sb2NhAwPRKxBTU1JQVxmbGFtajk1DAdj
ZXJ0cmVxMEwGC1sGAQQBgjcNAgExPjA8HiYAAQABAdABpAGYAAQABjAGEAdAB1
AFQAZQBtAHAAbABhAQZAR4SAFCAZQBIAFMAZQBYAHYAZQBYMHIGC1sGAQQBgjcN
AgIXZDBiAgEBH1oATQBpAGMACgBvAHMAbW8mAHQAIABSAFMAQAgAFMAQwBoAGEA
bgBuAGUAbAAGAEACgB5AHAAdABvAGCAGBhAHAAaABpAGMAIAABQAHIAbW82AGkA
ZAB1AHIAAQAwGAUGCSqGSIb3DQEJDDjGB1zCB1DAhBgkrBgEEAY13FAIEFB4SAFCA
ZQBIAFMAZQBYAHYAZQBYMBMGAA1UdJQQMMAoGCCsGAQUFBwMBMA4GA1UdDwEB/wQE
AwIFoDARBgNVHREEJDA1ghF3d3cuZmxhbW1pYS5sb2NhIINZmxhbW1pYS5sb2Nh
bDAdBgNVHQ4EFgQUQ3KEA+2V0WnQryrxbzqt3rCmGBkwDQYJKoZIhvcNAQEFBQAD
ggEBAADjsqD37bjWxdtLS9zVNu11UbXqd3K0r1k1jgCKwC0NwawpvcBPKRMR8Q
1S+b09uepc7hDZtN4TJxb3d0r1b1BN0T5U8L1Dg1mZefxib0dYfvthoC82FI9uIP
XnJZzM/UpmWcSuB+aSWHLJTyJYCIjqF2hXQY3hmpjSPfKLuH7+zjbQYJvRpp0T
tx8Aba51qme1A8h54C/SeLg1TYdUIGpe3Pq1iZug2+gr6UJ9kd0cLpJ1/j0AoQDz
oov0QEkjdPb+WcG1SC1jQ6NwqdeIxMy2yGr0/dCGYcWgWwH21knKYLQUvbKVEK8+
4uINFa0dQMUBJjRRNANfGzuIjNI-----
-----END NEW CERTIFICATE REQUEST-----

Certificate Template:

Web Server

Additional Attributes:

Attributes:

Submit >

After following the previous steps and downloading the certificate, I opened the command prompt and entered the command “certreq -accept certnew.cer” to install the new certificate. As stated previously, make sure you are in the correct directory.

```
C:\Users\flamj95.FLAMMIA\Desktop>certreq -accept certnew.cer
Installed Certificate:
Serial Number: 4e000000e40d7fb41ecc76bfe000000000000e
Subject: CN=www.flammia.local (DNS Name=www.flammia.local, DNS Name=flammia.local)
NotBefore: 4/21/2025 11:14 PM
NotAfter: 4/21/2027 11:14 PM
Thumbprint: b36a52d2a9dac2e98445ddd1dd67c083c761a801
```

From here, we can start to bind the certificate to the website we created. As pictured below, click on the website you want to bind the certificate to, and then click “Bindings” on the right side.

Connections

- Start Page
- SERVER3 (FLAMMIA\flamj95)
 - Application Pools
 - Sites
 - Default Web Site
 - TestPage

TestPage Home

Filter: Show All Group by: Area

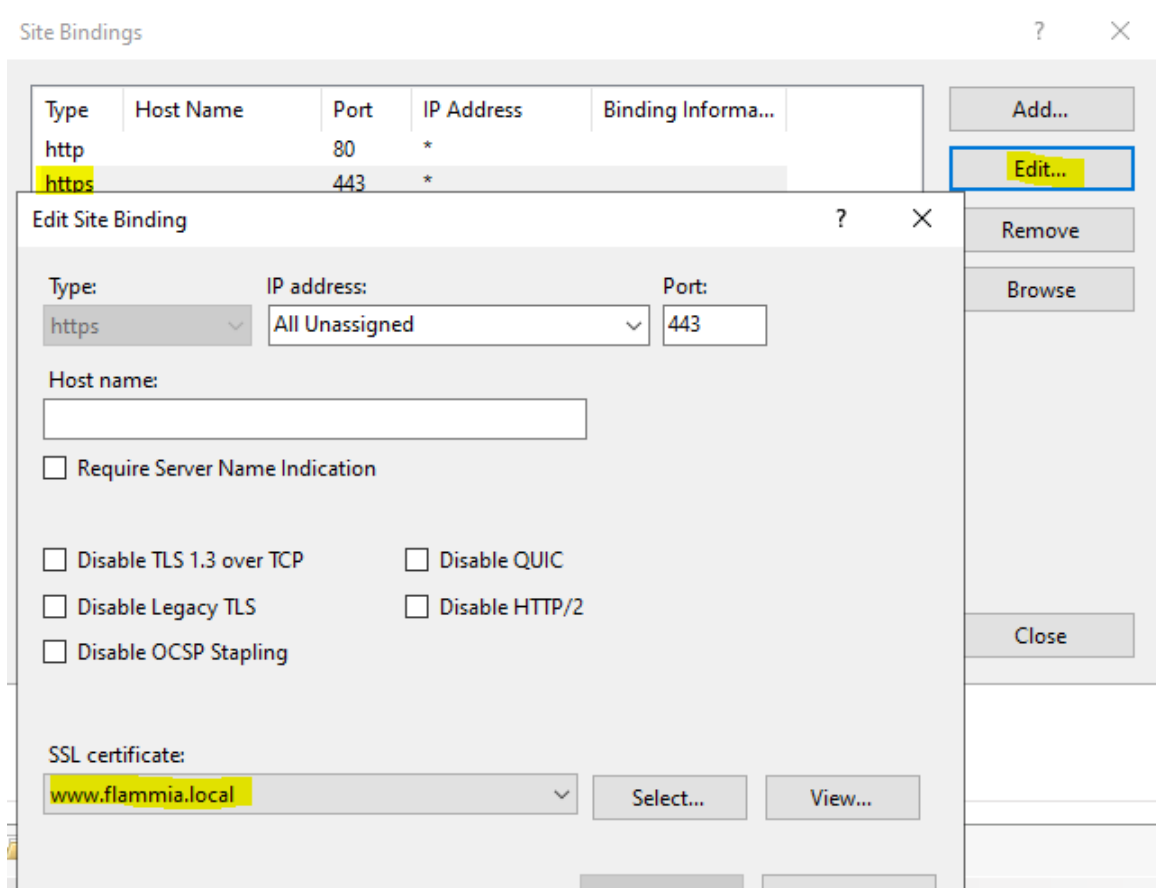
Authentic... Compression Default Document Directory Browsing Error Pages Handler Mappings HTTP Respon... Logging MIME Types Modules

Actions

- Explore
- Edit Permissions...
- Edit Site
 - Bindings...
 - Basic Settings...
- View Applications
- View Virtual Directories
- Manage Website

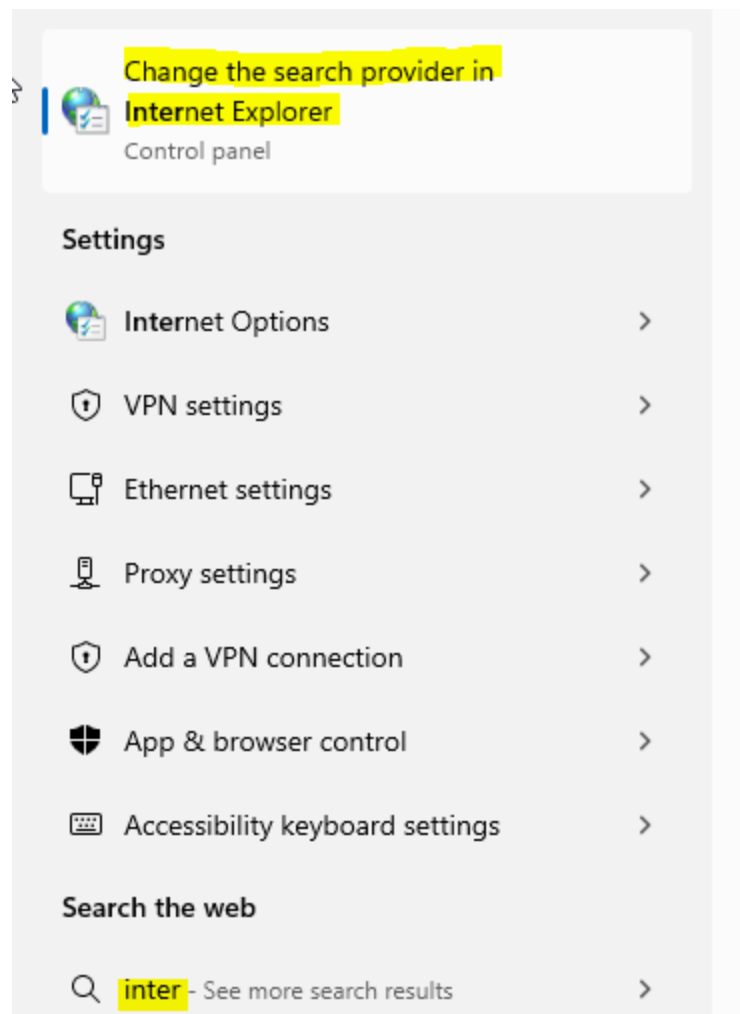
On the site bindings page, I clicked on the “https” type and then clicked edit.

From here, all we need to do is choose the SSL certificate we created. In my case, it was labeled as www.flammia.local. After choosing the correct certificate, click “OK” to finish.

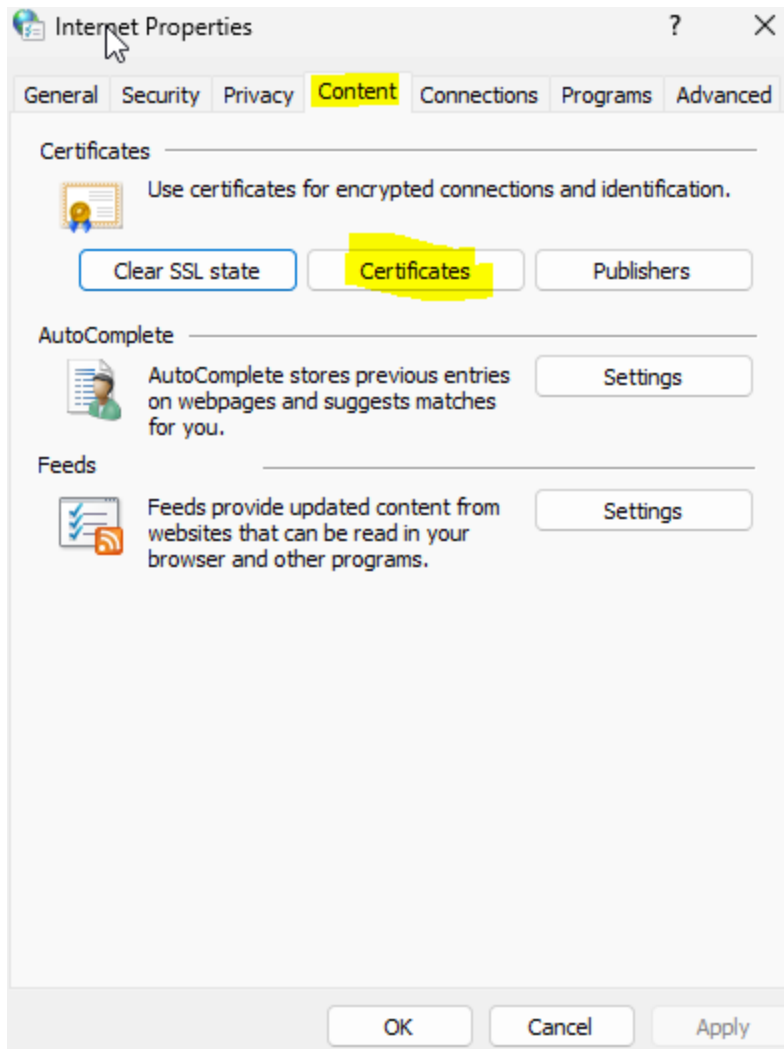


Finally, if everything has been configured correctly, we should be able to access www.flammia.local and see it as a secure site with a legitimate certificate with a computer in the domain.

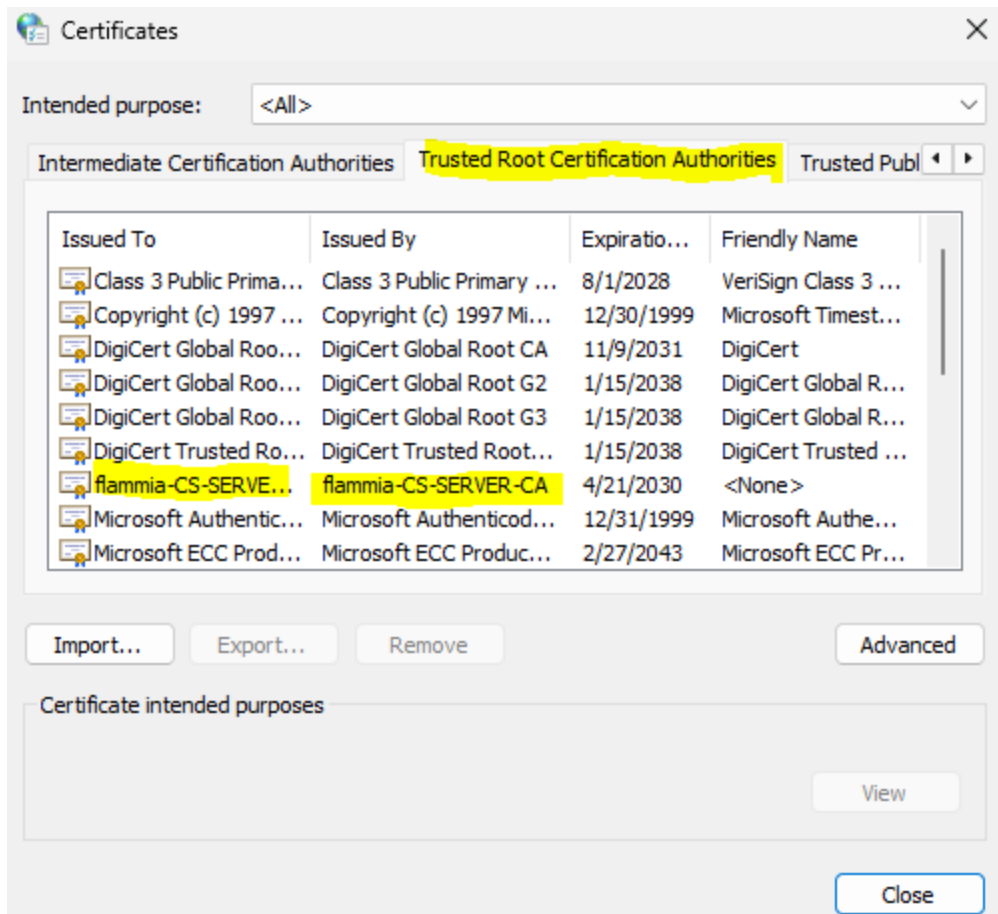
Before we test if our configurations are successful, I checked to see if our AD CS server is listed as a “Trusted Authority” on my Windows 11 VM. To do this, in the taskbar type in “inter” and click the “Change the search provider in Internet Explorer” option.



Once opened, click on the “content” tab, and then click on “Certificates” as highlighted below:

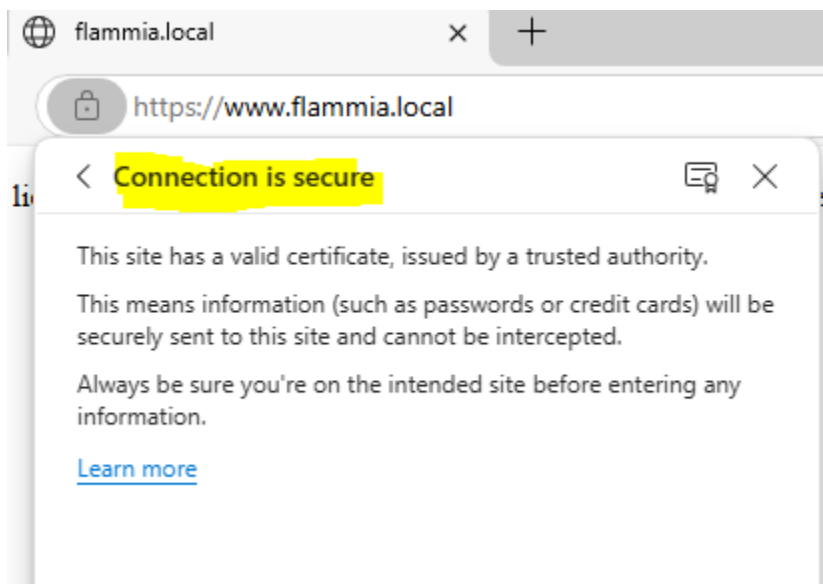
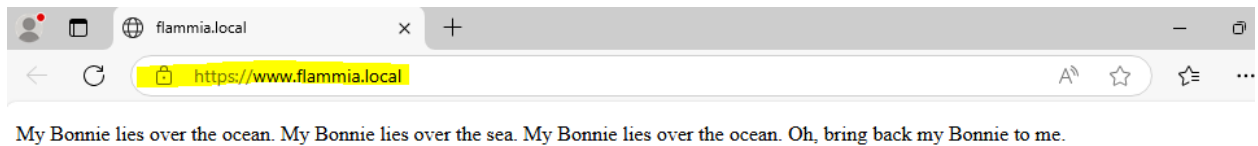


From this window, click on “Trusted Root Certification Authorities”. As you can see below, the name of the AD CS is on the trusted CA list.



Now to test if our certification works. To do this, I simply typed in www.flammia.local into the search browser.

As you can see below, I was able to access the website with no warnings, and the site is now labeled as “secure” due to the certificate.



CentOS Portion:

For this part of the lab, using Apache we will attempt to create a certificate request that can be fulfilled by the CS server.

To start, I opened a terminal and entered the command “openssl req -new -newkey rsa:2048 -nodes -keyout apache.key -out apache.csr” to create a private key called named “apache.key” and a certificate request named “apache.csr”.

```
[root@localhost ~]# openssl req -new -newkey rsa:2048 -nodes -keyout apache.key -out apache.csr
```

From here, we will be prompted to enter some basic info that will be included with the certificate. Most of the information included can be left blank, however the only part that needs to be filled in is the CN. This can be either the hostname or IP address of the VM. Here, I entered “192.168.206.128” as the CN.

```
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:
State or Province Name (full name) []:
Locality Name (eg, city) [Default City]:
Organization Name (eg, company) [Default Company Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:192.168.206.128
Email Address []:

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
```


If we enter the command “cat apache.csr”, we can view the certificate request like the ones created before on our IIS server.

```
[root@localhost ~]# cat apache.csr
-----BEGIN CERTIFICATE REQUEST-----
MIICoTCCAYkCAQAwxDELMAKGA1UEBhMCWFgxFTATBgNVBACMDERLZmF1bHQgQ2l0
eTEcMBoGA1UECgwTRGVmYXVsdCBDb2lwYW55IEx0ZDEYMBYGA1UEAwwPMtKyLjE2
OC4yMDYuMTI4MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA3gNvbDKF
WC/PTkn9nUbfwSnpNui0Zk6SxebcLH0+St7qbpX1MoDJBLTc8GxatDoAiunDLAbM
cJRTzXberXKMbPRY7lXvbtYL9upQ8f17E5A4KMuAGqSljPZQq6pYNM1HQnhpBtbk
9EpoXdLQzevDXN19I2+/wMRVP6Hq62IR0XoFWBfBEwR612IcZpION1mPdbBSSsMv
TmLY6EKubir/u4m9dj+T1oQIjjdzMRp3iLBQu54SJeAD/pzdrIC5p2YJzPXcdMDr
cCBgRv7U5sYq3z2U+CRRftUGP1eGqU3UcREnovnZxwhsjKTRdfKckChBjko06dKN
wqptJTJBQSNY4QIDAQABoAAwDQYJKoZIhvcNAQELBQADggEBAMkU2CazEjrQltcL
3WueMlf7uzMUK6f6mFedNgs3T8ZNkuzRlc9frAPHjd+WPUHUpU+VH7Ge1dx3XWRT
A0yz0foX5K4TUSjXgDr5MZqqIvtYV6ntL3KDFVp9/XsRUQ7/nH2oV7VNvoWYGc3K
RPD2cEwUjZVtB9GWmxqPtAuf0Ru4FCzV8cx9GMWCbgmRDsC8vR3C3aoN3bXajALB
uVXMbcmrX6y58kwbD1tLqQn7C6fcevZILR9xun3XJeKn82JiD2BjW0k5FIsmY+Cj
AWurGYSq2SwJNR3YQMEMHkHMR2mDCX4UC0eslEZTZiPbA0EjZMwMXhKq0tbIzaqs
onZFhms=
-----END CERTIFICATE REQUEST-----
```

From here, the process for completing the certification request will be very similar to before. In a web browser on the Linux machine, enter “(IPADDRESSHERE)/certsrv”.

Once on the main page of the site, click “Create a certificate request”, and then enter the certificate request found in the “apache.csr” file. After changing the template to “Web Server”, we can submit the request.

Microsoft Active Directory Certificate Services -- flammia-CS-SERVER-CA

Submit a Certificate Request or Renewal Request

To submit a saved request to the CA, paste a base-64-encoded CMC or PKCS #10 certificate request or PKCS #

Saved Request:

Base-64-encoded certificate request (CMC or PKCS #10 or PKCS #7):	-----BEGIN CERTIFICATE REQUEST----- MIICoTCCAYKCAQAwXDELMAkGA1UEBhMCWFgxFtATB eTEcMBoGA1UECgwTRGVmYXVsdCBDb21wYW55IEExO OC4yMDYuMTI4MIIIBIjANBgkqhkiG9w0BAQEFAAOC PCENAWkGGDsGq2dS90b/aoMV5bg2Q0+Bz/EJ7Y+Sh
---	---

Certificate Template:

Web Server

Additional Attributes:

Attributes:

Submit >

After downloading the certification, we will need to move the private key and the certificate inside of the “tls” directory.

First, I entered the command “mv centos.key /etc/pki/tls/private” to move the private key to the correct directory.

```
[root@localhost ~]# mv apache.key /etc/pki/tls/private
```

Next, I entered the command “mv apachecert.cer /etc/pki/tls/certs” to move the certificate to the correct directory.

```
[root@localhost Downloads]# mv apachecert.cer /etc/pki/tls/certs
```

For the last step, we will need to edit the “ssl.conf” file found in the “conf.d” directory.

Using nano, enter the command “nano /etc/httpd/conf.d/ssl.conf” to start editing the file. From here, scroll down until you find the “SSLCertificateFile” and the “SSLCertificateKeyFile”. By default, the files are pointing towards the localhost files. All we need to do is change the names of the files to the ones we created.

As pictured below, I changed “localhost.cer” to “apachecert.cer”, and then changed “localhost.key” to “apache.key”.

```
GNU nano 5.6.1 /etc/httpd/conf.d/ssl.conf
# The OpenSSL system profile is used by default. See
# update-crypto-policies(8) for more details.
#SSLProtocol all -SSLv3
#SSLProxyProtocol all -SSLv3

# User agents such as web browsers are not configured for the user's
# own preference of either security or performance, therefore this
# must be the prerogative of the web server administrator who manages
# cpu load versus confidentiality, so enforce the server's cipher order.
SSLHonorCipherOrder on

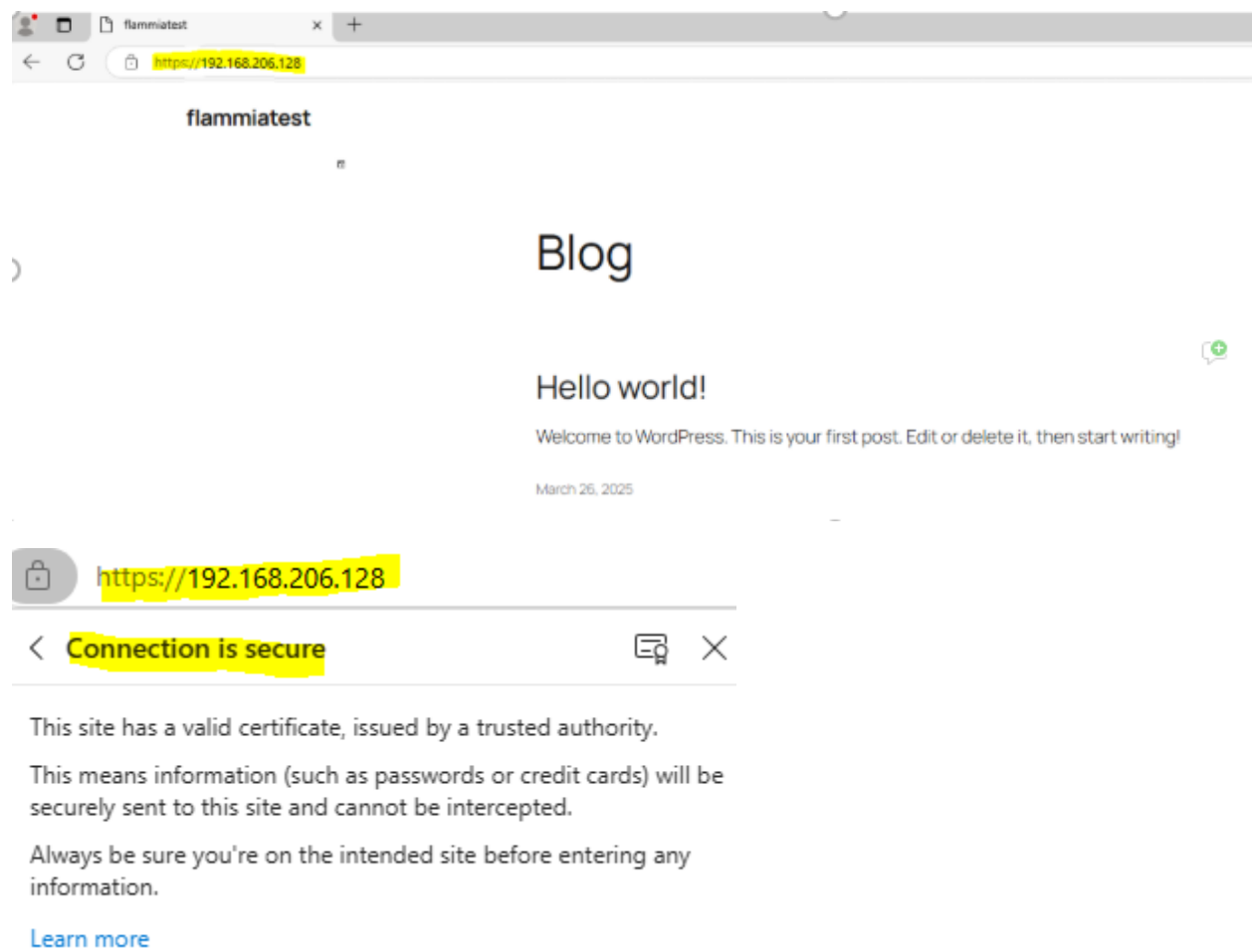
# SSL Cipher Suite:
# List the ciphers that the client is permitted to negotiate.
# See the mod_ssl documentation for a complete list.
# The OpenSSL system profile is configured by default. See
# update-crypto-policies(8) for more details.
SSLCipherSuite PROFILE=SYSTEM
SSLProxyCipherSuite PROFILE=SYSTEM

# Point SSLCertificateFile at a PEM encoded certificate. If
# the certificate is encrypted, then you will be prompted for a
# pass phrase. Note that restarting httpd will prompt again. Keep
# in mind that if you have both an RSA and a DSA certificate you
# can configure both in parallel (to also allow the use of DSA
# ciphers, etc.)
# Some ECC cipher suites (http://www.ietf.org/rfc/rfc4492.txt)
# require an ECC certificate which can also be configured in
# parallel.
SSLCertificateFile /etc/pki/tls/certs/apachecert.cer

# Server Private Key:
# If the key is not combined with the certificate, use this
# directive to point at the key file. Keep in mind that if
# you've both a RSA and a DSA private key you can configure
# both in parallel (to also allow the use of DSA ciphers, etc.)
# ECC keys, when in use, can also be configured in parallel
SSLCertificateKeyFile /etc/pki/tls/private/apache.key
```

Finally, we can test if our configurations were correct. Here, I switched over to my Windows 11 VM and entered “192.168.206.128” into my browser.

As you can see below, I was able to access the webpage with no warnings, and the connection now appears secure.



Additionally, by checking the certificate's details, we can see that the certificate was issued by our CS Server.

Certificate Viewer: 192.168.206.128

General

Details

Issued To

Common Name (CN)	192.168.206.128
Organization (O)	Default Company Ltd
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	flammia-CS-SERVER-CA
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>